# 2.61 Moments: Ladders and Hinges

Worked Solutions

**1 a)**

Distance: $2\sin 30° = 1$

M($O$) $R(\circlearrowleft)$: $15 \times 1 = 15$ Nm

15 Nm anti-clockwise

**1 b)**

Distance: $2\sin 75°$

M($O$) $R(\circlearrowleft)$: $-5 \times 2\sin 75° = -9.66$ Nm

9.66 Nm clockwise (3 SF)

**1 c)**

Distance: $10\sin 25°$

M($O$) $R(\circlearrowleft)$: $-5 \times 10\sin 25° = -21.1$ Nm

21.1 Nm clockwise (3 SF)

**1 d)**

Distance: $3.5\sin 60°$

M($O$) $R(\circlearrowleft)$: $4 \times 3.5\sin 60° = 12.1$ Nm

12.1 Nm anti-clockwise (3 SF)

**1 e)**

M($O$) $R(\circlearrowleft)$: $2 \times 3\sin 20° - 6 \times 3\sin 45°$

$= -10.7$ Nm

10.7 Nm clockwise (3 SF)

**1 f)**

M($O$) $R(\circlearrowleft)$: $12 \times 5\sin 45° + 15 \times 6\sin 40°$

$= 100$ Nm

100 Nm anti-clockwise (3 SF)

**1 g)**

M($O$) $R(\circlearrowleft)$: $-3 \times 4\sin 65° - 2 \times 3\sin 45°$

$= -15.1$ Nm

15.1 Nm clockwise (3 SF)

**1 h)**

M($O$) $R(\circlearrowleft)$: $40 \times 3\sin 70° + 55 \times 10\sin 25°$

$= 345$ Nm

345 Nm anti-clockwise (3 SF)

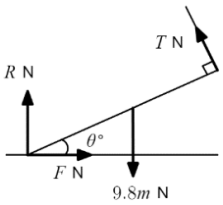
**1 i)**

M($O$) $R(\circlearrowleft)$: $\quad 6 \times 4\sin 40° - 8 \times 4\sin 75°$

$= -15.5$ Nm

15.5 Nm clockwise (3 SF)

**2 a)**




**2 b)**

Weight: $9.8 \times 30 = 294$ N

M($A$) $R(\circlearrowleft)$: $\quad -294\cos 30° \times 5 + 10T = 0$

$\therefore T = 127$ N (3 SF)

$T = 130$ N (2 SF – AQA)

$R(\rightarrow)$: $F - T\cos 60° = 0 \Rightarrow F = 63.65286718\ldots$

$R(\uparrow)$: $R - 294 + T\sin 60° = 0 \Rightarrow R = 183.75$

$\mu = 0.346$ (3 SF)

$\mu = 0.35$ (2 SF – AQA)

**2 c)**

M($A$) $R(\circlearrowleft)$: $\quad -9.8m\cos 20° \times 4 + 8 \times 180 = 0$

$\Rightarrow m = 39.1$ kg (3 SF)

$m = 40$ kg (2 SF – AQA)

$R(\rightarrow)$: $F - 180\cos 70° = 0 \Rightarrow F = 61.5636258\ldots$

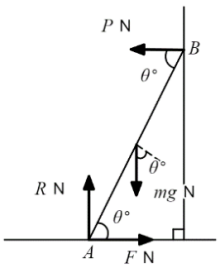
$R(\uparrow)$: $R - 39.1 \times 9.8 + 180\sin 70° = 0$

$\Rightarrow R = 213.9593263\ldots$

$\mu = 0.288$ (3 SF)

$\mu = 0.29$ (2 SF – AQA)

**3 a)**




**3 b)**

$R(\uparrow)$: $R - 10 \times 9.8 = 0 \Rightarrow R = 98$

M$(B)$ $R(\circlearrowleft)$:

$98\cos 60° \times 2.5 - 98\sin 30° \times 5 + F\sin 60° \times 5 = 0$

$F = 28.29016319$

$\mu = \frac{F}{R} = 0.289$ (3 SF)

$\mu = 0.29$ (2 SF – AQA)

**3 c)**

$R(\uparrow)$: $R - 10 \times 9.8 = 0 \Rightarrow R = 98$

M$(B)$ $R(\circlearrowleft)$:

$98\cos 45° \times 2.5 - 98\sin 45° \times 5 + F\sin 45° \times 5 = 0$

$F = 49$

$\mu = \frac{F}{R} = 0.5$

**3 d)**

$R(\uparrow)$: $R - 9.8M = 0 \Rightarrow R = 9.8M$

M$(B)$ $R(\circlearrowleft)$:

$9.8M\cos 60° \times \frac{a}{2} - 9.8M\sin 30° \times a + F\sin 60° \times a = 0$

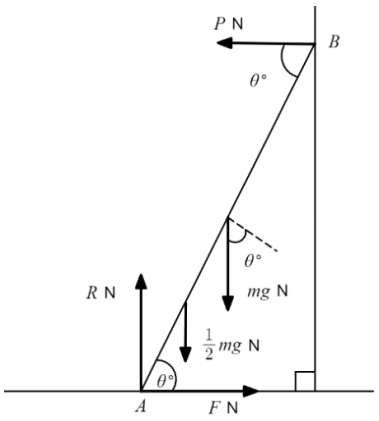
$-\frac{49}{20}M + F\sin 60° = 0$

$F = \frac{49\sqrt{3}}{30}M$

$\mu = \frac{F}{R} = 0.289$ (3 SF)

$\mu = 0.29$ (2 SF – AQA)

**3 e)**




$R(\uparrow)$: $R - 9.8m - 4.9m = 0 \Rightarrow R = 14.7m$

M$(B)$ $R(\circlearrowleft)$:

$9.8m\cos 60° \times \frac{a}{2} + 4.9m\cos 60° \times \frac{3a}{4} - 14.7m\sin 30° \times a + F\sin 60° \times a = 0$

$-\frac{49}{16}m + F\sin 60° = 0$

$F = \frac{49}{16\sin 60°}m$

$\mu = \frac{F}{R} = 0.241$ (3 SF)

$\mu = 0.24$ (2 SF – AQA)

**3 f)**

$R(\uparrow)$: $R - 9.8m - 4.9m = 0 \Rightarrow R = 14.7m$

M$(B)$ $R(\circlearrowleft)$:

$9.8m\cos 45° \times \frac{a}{2} + 4.9m\cos 45° \times \frac{3a}{4} - 14.7m\sin 45° \times a + F\sin 45° \times a = 0$

$\frac{49\sqrt{2}}{20}m + \frac{147\sqrt{2}}{80}m - \frac{147\sqrt{2}}{20}m + \frac{\sqrt{2}}{2}F = 0$

$-\frac{49}{16}m + \frac{1}{2}F = 0$

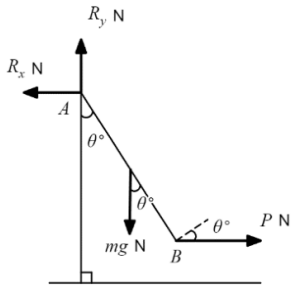
$F = \frac{49}{8}m$

$\mu = \frac{F}{R} = \frac{5}{12} = 0.417$ (3 SF)

$\mu = 0.042$ (2 SF – AQA)

**4 a)**

The reaction at the wall can be split into horizontal and vertical components, which must balance the system:




**4 b)**

M($A$) $R(\circlearrowleft)$:

$-2 \times 9.8 \cos 30° \times 5 + P \cos 30° \times 10 = 0$

$P = 9.8$ N

Resolving horizontally and vertically:

$R_x = 9.8$ N

$R_y = 2 \times 9.8 = 19.6$ N

$|R| = \sqrt{9.8^2 + 19.6^2} = 21.9$ N (3 SF)

$|R| = 22$ N (2 SF – AQA)

**4 c)**

M($A$) $R(\circlearrowleft)$:

$-5 \times 9.8 \cos 45° \times \frac{a}{2} + P \cos 45° \times a = 0$

$P = 24.5$ N

Resolving horizontally and vertically:

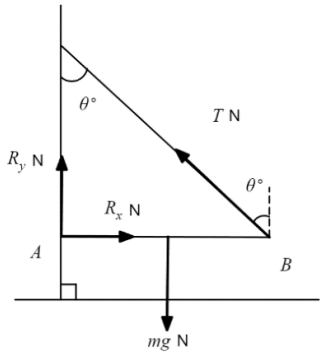
$R_x = 24.5$ N

$R_y = 5 \times 9.8 = 49$ N

$|R| = \sqrt{24.5^2 + 49^2} = 54.8$ N (3 SF)

$|R| = 55$ N (2 SF – AQA)

**5 a)**




**5 b)**

M($A$) $R(\circlearrowleft)$: $T\cos 30° \times 10 - 3 \times 9.8 \times 5 = 0$

$\Rightarrow T = \frac{49}{5}\sqrt{3} = 17.0$ N (3 SF)

$T = 17$ N (2 SF – AQA)

$R(\uparrow)$: $R_y - 3 \times 9.8 + T\cos 30° = 0$

$\Rightarrow R_y = 14.7$ N

$R(\rightarrow)$: $R_x - T\sin 30° = 0$

$\Rightarrow R_x = \frac{49}{10}\sqrt{3}$ N

$|R| = \sqrt{14.7^2 + \left(\frac{49}{10}\sqrt{3}\right)^2} = \frac{49}{5}\sqrt{3} = 17.0$ N (3 SF)

$|R| = 17$ N (2 SF – AQA)

**5 c)**

M($A$) $R(\circlearrowleft)$: $T\cos 60° \times a - 15 \times 9.8 \times \frac{a}{2} = 0$

$\Rightarrow T = 147$ N

$T = 150$ N (2 SF – AQA)

$R(\uparrow)$: $R_y - 15 \times 9.8 + T\cos 60° = 0$

$\Rightarrow R_y = 73.5$ N

$R(\rightarrow)$: $R_x - T\sin 60° = 0$

$\Rightarrow R_x = \frac{147}{2}\sqrt{3}$ N

$|R| = \sqrt{73.5^2 + \left(\frac{147}{2}\sqrt{3}\right)^2} = 147$ N (3 SF)

$|R| = 150$ N (2 SF – AQA)

**5 d)**

M($A$) $R(\circlearrowleft)$: $T\cos 45° \times a - 10 \times 9.8 \times \frac{a}{2} = 0$

$\Rightarrow T = 49\sqrt{2} = 69.3$ N (3 SF)

$T = 69$ N (2 SF – AQA)

$R(\uparrow)$: $R_y - 10 \times 9.8 + T\cos 45° = 0$

$\Rightarrow R_y = 49$ N

$R(\rightarrow)$: $R_x - T\sin 45° = 0$

$\Rightarrow R_x = 49$ N

$|R| = \sqrt{49^2 + 49^2} = 49\sqrt{2} = 69.3$ N (3 SF)

$|R| = 69$ N (2 SF – AQA)

**5 e)**

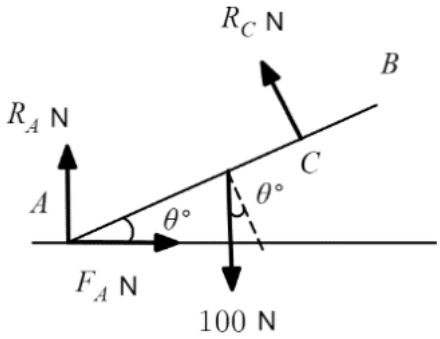
M($A$) $R(\circlearrowleft)$:

$T\cos 40° \times 8 - 5 \times 9.8 \times 4 - 3 \times 9.8 \times 2 = 0$

$\Rightarrow T = 41.6$ N (3 SF)

$\Rightarrow T = 42$ N (2 SF – AQA)

**6 a)**




**6 b)**

$\tan\theta = \frac{7}{24}, \sin\theta = \frac{7}{25}, \cos\theta = \frac{24}{25}$

M($A$) $R(\circlearrowleft)$: $R_C \times 4a - 100 \times \frac{24}{25} \times 2.5a = 0$

$R_C = 60$

$R(\uparrow)$: $60 \times \frac{24}{25} + R_A - 100 = 0$

$\therefore R_A = 42.4$ N

$R_A = 40$ N (3 SF – AQA)

$R(\rightarrow)$: $F_A - 60 \times \frac{7}{25} = 0$

$\therefore F_A = 16.8$ N (3 SF)

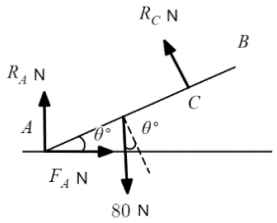
$F_A = 17$ N (2 SF – AQA)

**6 c)**

$\mu = \frac{F_A}{R_A} = \frac{21}{53} = 0.396$ (3 SF)

$\mu = 0.40$ (2 SF – AQA)

**7 a)**




**7 b)**

$\tan\theta = \frac{7}{24}, \sin\theta = \frac{7}{25}, \cos\theta = \frac{24}{25}$

M($A$) $R(\circlearrowleft)$: $R_C \times 4a - 80 \times \frac{24}{25} \times 2a = 0$

$R_C = 38.4$

$R(\uparrow)$: $38.4 \times \frac{24}{25} + R_A - 80 = 0$

$\therefore R_A = 43.136$ N

$R_A = 43.1$ N (3 SF)

$R_A = 43$ N (2 SF – AQA)

$R(\rightarrow)$: $F_A - 38.4 \times \frac{7}{25} = 0$

$\therefore F_A = 10.752$ N

$F_A = 10.8$ N (3 SF)

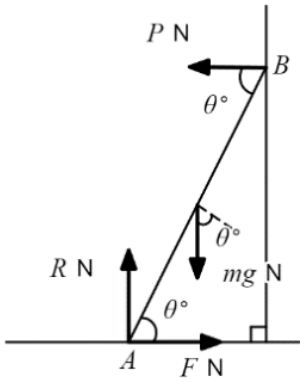
$F_A = 11$ N (3 SF – AQA)

**7 c)**

$\mu = \frac{F_A}{R_A} = \frac{84}{337} = 0.249$ (3 SF)

$\mu = 0.25$ (2 SF – AQA)

**8 a)**




M($A$) $R(\circlearrowleft)$:

$-9.8m \times \cos\theta \times \frac{a}{2} + P \times \sin\theta \times a = 0$

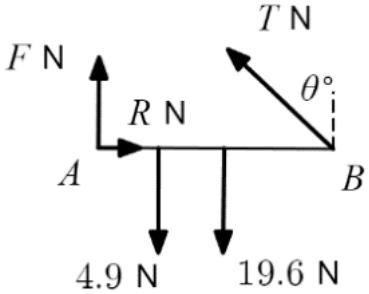
$P = \frac{4.9ma\cos\theta}{a\sin\theta} = \frac{4.9m}{\tan\theta}$

$R(\uparrow)$: $R - 9.8m = 0 \Rightarrow R = 9.8m$

$R(\rightarrow)$: $F - \frac{4.9m}{\tan\theta} = 0 \Rightarrow F = \frac{4.9m}{\tan\theta}$

$\therefore \mu = \frac{\left(\frac{4.9m}{\tan\theta}\right)}{9.8m} = \frac{1}{2\tan\theta} \Rightarrow 2\mu\tan\theta = 1$ as req.

**9 a)**




$\tan\theta = \frac{3}{4}, \sin\theta = \frac{3}{5}, \cos\theta = \frac{4}{5}$

M($A$) $R(\circlearrowleft)$:

$\frac{4}{5}T \times 4a - 19.6 \times 2a - 4.9a = 0$

$\Rightarrow T = \frac{441}{32} = 13.8$ N (3 SF)

$T = 14$ N (2 SF – AQA)

**9 b)**

$R(\uparrow)$: $F + \frac{441}{32} \times \frac{4}{5} - 4.9 - 19.6 = 0 \Rightarrow F = 13.475$ N

$R(\rightarrow)$: $R - \frac{441}{32} \times \frac{3}{5} = 0 \Rightarrow R = \frac{1323}{160} = 82.7$ N (3 SF)

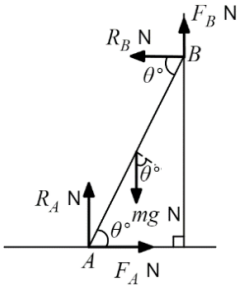
$R = 22$ N (2 SF – AQA)

**9 c)**

$\mu = \frac{44}{27} = 1.63$ (3 SF)

$\mu = 1.6$ (2 SF – AQA)

**10)**




**10 a)**

$F_B = 0.1R_B$ and $F_A = 0.4R_A$

$R(\rightarrow)$: $0.4R_A - R_B = 0 \Rightarrow 0.4R_A = R_B$

$\therefore R_A = 2.5R_B$ therefore the normal reaction at the floor is 2.5 times larger than the normal reaction at the wall

**10 b)**

$R_A = 15$ so $R_B = 6$

$R(\uparrow)$: $15 - mg + F_B = 0$

$\Rightarrow mg = 15 + 0.1 \times 6 \therefore m = \frac{78}{5g}$

**10 c)**

M($A$) $R(\circlearrowleft)$:

$-amg\cos\theta + 2a \times 6\sin\theta + 2a \times 0.6\cos\theta = 0$

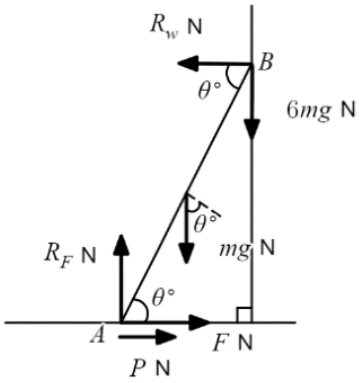
$-mg\cos\theta + 12\sin\theta + 1.2\cos\theta = 0$

$-\frac{78}{5}\cos\theta + 12\sin\theta + 1.2\cos\theta = 0$

$12\sin\theta = 14.4\cos\theta$

$\tan\theta = \frac{6}{5} \therefore \theta = 50.2°$ (3 SF)

**11 a)**




$\tan\theta = \frac{24}{7}, \sin\theta = \frac{24}{25}, \cos\theta = \frac{7}{25}$

M($A$) $R(\circlearrowleft)$:

$-\frac{7}{25}mg \times 3a + \frac{24}{25}R_W \times 6a - \frac{7}{25} \times 6mg \times 6a = 0$

$R_W = \frac{91}{48}mg$ as req

**11 b)**

$R(\uparrow)$: $R_F - 6mg - mg = 0 \Rightarrow R_F = 7mg$

$F = 0.25 \times 7mg = \frac{7}{4}mg$

$R(\rightarrow)$: $F + P - R_W = 0$

$\frac{7}{4}mg + P - \frac{91}{48}mg = 0 \therefore P = \frac{7}{48}mg$

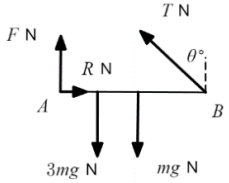
Or by reversing the direction of $F$

$R(\rightarrow)$: $-F + P - R_W = 0$

$-\frac{7}{4}mg + P - \frac{91}{48}mg = 0 \therefore P = \frac{175}{48}mg$

$\frac{7}{48}mg \leq P \leq \frac{175}{48}mg$

**12 a)**




$\tan\theta = \frac{4}{3}, \sin\theta = \frac{4}{5}, \cos\theta = \frac{3}{5}$

M($A$) $R(\circlearrowleft)$:

$-5mg \times x - mg \times \frac{1}{2}a + \frac{3}{5}T \times a = 0$

$T = \frac{5mgx + \frac{1}{2}mga}{\frac{3}{5}a} = \frac{mg\left(5x + \frac{1}{2}a\right)}{\frac{3}{5}a} = \frac{5mg(10x+a)}{6a}$ as req

**12 b)**

$8mg = \frac{5mg(10x+a)}{6a}$

$\Rightarrow 48mga = 50mgx + 5mga$

$\Rightarrow 43a = 50x$

$x = \frac{43}{50}a$ so 86% across from $A$